

SE LAB-01

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Use-Case Flow Specifications (Multi-Use Case Package)

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #07 — Campus & Academic Operations

UC-03: Update Ticket Status (FR-003)

Primary Actor: Maintenance Staff / Warden	Supporting Actors: Hostel Resident
Preconditions: Ticket is in 'Assigned' or 'In Progress' state.	Postconditions (Success): Status updated, timestamped, resident notified.

Main Success Scenario

- Assigned Maintenance Staff or Warden accesses ticket details via mobile app or portal.
- User selects new status update (e.g., transition from 'Open' → 'In Progress' upon arrival, or 'In Progress' → 'Resolved' after repair).
- User optionally attaches work progress notes or photo of completed work.
- User submits status update.
- System validates transition logic, records server timestamp, and updates ticket status.
- System sends automated status notification alert to reporting resident's dashboard and mobile app within 1 minute. Use case ends.

Alternate Flow — 2a: Spare Parts Delay

- 2a. Triggered at step 2 if repair requires specialized spare parts not currently in stock.
- 2a1. Staff sets status to 'On Hold — Awaiting Parts' and enters estimated delay.
- 2a2. System updates resident dashboard with delay explanation and revised resolution target. Use case ends.

Requirements_Table

	A	B	C	D	E	F
1	Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
2	FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity (Plumbing/Electrical/Carpentry; Low/Medium/High).	High	Pass: Ticket is created, assigned a unique ID, and routed to the warden queue. Fail: Incomplete form (e.g., missing room tag) is accepted.	Core intake function; accurate tagging and severity are required to route and prioritize the ticket.
3	FR-002	Functional	The system shall allow the Maintenance Warden to assign a submitted ticket to a specific maintenance staff member and set a target resolution SLA based on issue severity.	High	Pass: Warden selects a staff member from a list, ticket status changes to "Assigned," and the assigned staff is notified within 1 minute. Fail: Ticket remains unassigned after the warden submits the assignment.	Ensures every logged issue is routed to a responsible person for timely resolution.
4	FR-003	Functional	The system shall allow the Maintenance Warden or assigned staff to update a ticket's status (Open → In Progress → Resolved) as work progresses.	High	Pass: Each status change is timestamped and visible on the resident's ticket view within 1 minute. Fail: A status change is not reflected to the resident.	Keeps residents informed and creates an auditable timeline needed for SLA tracking.
5	FR-004	Functional	The system shall require the reporting resident to confirm and record a closure sign-off before a ticket can be marked "Closed."	Medium	Pass: A ticket cannot transition to Closed without a stored sign-off (resident ID + timestamp). Fail: A ticket is marked Closed with no sign-off record.	Verifies the issue is actually resolved from the resident's perspective and creates accountability for maintenance staff.
6	FR-005	Functional	The system shall allow residents to view the real-time status and history of all maintenance tickets they have submitted.	Medium	Pass: The resident's dashboard lists every submitted ticket with current status and last-updated timestamp. Fail: A submitted ticket is missing from the resident's list.	Transparency reduces duplicate ticket submissions and manual status-check requests to the hostel office.
7	NFR-001	Nonfunctional — Performance & Security	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without a status change.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load, and escalation alerts fire within the 24-hour window.	Prevents high-severity issues (e.g., electrical hazards) from being silently ignored past the SLA.
8	NFR-002	Nonfunctional — Usability	The system shall allow a resident to complete and submit a new ticket, including a photo attachment, within 90 seconds on a standard smartphone browser.	Medium	Pass: Usability testing with 10 residents shows average submission time ≤ 90s. Fail: average exceeds 90s.	Low reporting friction increases adoption and reduces issues going unreported.

UseCase Diagram

UML Use-Case Diagram: Hostel Maintenance & Issue Ticketing System

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